

ANSWER KEY

GATE EC — GATE EC 2026 Trend-Based Mock Paper
2

Q.No.	Type	Answer	Marks
1	2MCQ	—	2
2	2MCQ	A	2
3	2NAT	—	2
4	2MCQ	—	2
5	2NAT	B	2
6	2MCQ	—	2
7	2MCQ	—	2
8	2NAT	B	2
9	2NAT	—	2
10	2MCQ	—	2
11	2NAT	B	2
12	2MCQ	B	2
13	2MCQ	B	2
14	1MCQ	B	1
15	2MCQ	—	2
16	2MSQ	—	2
17	2MCQ	—	2
18	2MCQ	—	2
19	2NAT	—	2
20	2MCQ	—	2
21	2MCQ	—	2
22	2MCQ	—	2
23	1NAT	B	1
24	1NAT	B	1
25	1MCQ	B	1
26	1MCQ	B	1
27	1MCQ	A	1
28	1NAT	C	1
29	1MCQ	C	1
30	1MCQ	B	1
31	1MCQ	B	1
32	1NAT	A	1
33	1NAT	A	1
34	1NAT	B	1
35	1MCQ	A	1
36	1NAT	A	1

37	1MCQ	B	1
38	1MCQ	B	1
39	1MCQ	A	1
40	1MCQ	A	1
41	1MCQ	A	1
42	1MCQ	A	1
43	1MCQ	A	1
44	1MCQ	A	1
45	1NAT	B	1
46	1MCQ	D	1
47	1MCQ	C	1
48	1MCQ	A	1
49	2MSQ	—	2
50	2MCQ	—	2
51	2MCQ	—	2
52	2MCQ	—	2
53	2MCQ	—	2
54	2NAT	—	2
55	2MCQ	—	2
56	2MCQ	—	2
57	2NAT	—	2
58	1MCQ	B	1
59	2NAT	—	2
60	2MCQ	—	2
61	2MCQ	—	2
62	2NAT	—	2
63	2MCQ	—	2
64	1MCQ	—	1
65	1MCQ	—	1

Q1. General Aptitude | 2 marks | 2MCQ
For a real number > 1 , $1 \ 1 \ 1 \log 2 + \log 3 + \log 4 = 1$ The value of is

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 7 Q4

Q2. General Aptitude | 2 marks | 2MCQ
Choose opposite of 'benevolent':

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q10

Q3. General Aptitude | 2 marks | 2NAT
P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms....

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 7 Q2

Q4. General Aptitude | 2 marks | 2MCQ
If `` denotes increasing order of intensity, then the meaning of the words [charm enamor bewitch] is analogous to [bored....

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 7 Q1

Q5. General Aptitude | 2 marks | 2NAT
If train 100m crosses pole in 10s, speed:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q10

Q6. General Aptitude | 2 marks | 2MCQ
Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify th...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q1

Q7. General Aptitude | 2 marks | 2MCQ
The 12 musical notes are given as , #, , #, , , #, , #, , #, and . Frequency of each note is 122 times the frequency of ...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q3

Q8. General Aptitude | 2 marks | 2NAT
Sum of first 100 natural numbers:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q10

Q9. General Aptitude | 2 marks | 2NAT

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated ...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q4

Q10. General Aptitude | 2 marks | 2MCQ

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above ...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q2

Q11. General Aptitude | 2 marks | 2NAT

Choose synonym of 'trivial':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q10

Q12. Engineering Mathematics | 2 marks | 2MCQ

$i^4 =$

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q1

Q13. Engineering Mathematics | 2 marks | 2MCQ

Integral of $1/x$ dx:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q1

Q14. Engineering Mathematics | 1 marks | 1MCQ

$P(A \cap B) = ?$

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q1

Q15. Network, Signals & Systems | 2 marks | 2MCQ

For a causal discrete-time LTI system with transfer function $H(z) = (2z^2 + 3z + 13)/(z^2 - 13z + 22)$ - which of the following statements ...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q24

Q16. Network, Signals & Systems | 2 marks | 2MSQ

In the circuit given below, the switch S was kept open for a sufficiently long time and is closed at time $t = 0$. The time ...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q32

Q17. Network, Signals & Systems | 2 marks | 2MCQ

The intrinsic carrier concentration of a semiconductor is $2.5 \times 10^{16} \text{ /m}^3$ at 300 K. If the electron and hole mobilities a...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q46

Q18. Network, Signals & Systems | 2 marks | 2MCQ
Consider the following series: (i) =1 1 (ii) =1 1 (+1) (iii) =1 1 ! Choose the correct option.

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q12

Q19. Network, Signals & Systems | 2 marks | 2NAT
The propagation delay of the 2 $\times$ 1 MUX shown in the circuit is 10 ns. Consider the propagation delay of the inverter as ...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q41

Q20. Network, Signals & Systems | 2 marks | 2MCQ
An ideal p-n junction germanium diode has a reverse saturation current of 10 μ A at 300 K. The voltage (in Volts, rounded...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q63

Q21. Network, Signals & Systems | 2 marks | 2MCQ
An NMOS transistor operating in the linear region has of 5 A at of 0.1 V. Keeping constant, the is increased to 1.5 V. G...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q64

Q22. Network, Signals & Systems | 2 marks | 2MCQ
Consider the discrete-time system below with input $x[n]$ and output $y[n]$. In the figure, $1[n]$ and $2[n]$ denote the impulse respo...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q17

Q23. Electronic Devices | 1 marks | 1NAT
Zener diode used for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q9

Q24. Electronic Devices | 1 marks | 1NAT
SCR has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q9

Q25. Electronic Devices | 1 marks | 1MCQ
MOSFET is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q8

Q26. Electronic Devices | 1 marks | 1MCQ
n-type majority carriers:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q3

Q27. Analog Circuits | 1 marks | 1MCQ
RC oscillator frequency:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q9

Q28. Analog Circuits | 1 marks | 1NAT
Ideal op-amp open-loop gain:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 EC Session 1 Q2

Q29. Analog Circuits | 1 marks | 1MCQ
Common emitter has:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 EC Session 1 Q3

Q30. Analog Circuits | 1 marks | 1MCQ
Common emitter has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q4

Q31. Analog Circuits | 1 marks | 1MCQ
FET is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q3

Q32. Digital Circuits | 1 marks | 1NAT
ROM is:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q4

Q33. Digital Circuits | 1 marks | 1NAT
4-bit binary counter counts:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q4

Q34. Digital Circuits | 1 marks | 1NAT
D flip-flop follows input after:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q5

Q35. Digital Circuits | 1 marks | 1MCQ
TTL stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q3

Q36. Control Systems | 1 marks | 1NAT
Routh-Hurwitz determines:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q7

Q37. Control Systems | 1 marks | 1MCQ
Transfer function is defined:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q5

Q38. Control Systems | 1 marks | 1MCQ
Lead compensator:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q6

Q39. Control Systems | 1 marks | 1MCQ
Nyquist criterion uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q6

Q40. Communication Systems | 1 marks | 1MCQ
PCM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q4

Q41. Communication Systems | 1 marks | 1MCQ
AM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q6

Q42. Communication Systems | 1 marks | 1MCQ
FM has:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q5

Q43. Communication Systems | 1 marks | 1MCQ
ASK modulation varies:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q5

Q44. Electromagnetics | 1 marks | 1MCQ
Characteristic impedance $Z_0 =$:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q7

Q45. Electromagnetics | 1 marks | 1NAT
Waveguide supports:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q7

Q46. Electromagnetics | 1 marks | 1MCQ
Skin depth increases with:

Answer: D

Correct Answer: D

Source: Adapted from GATE 2023 EC Session 1 Q6

Q47. Electromagnetics | 1 marks | 1MCQ
Maxwell has how many equations:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 EC Session 1 Q8

Q48. Control Systems | 1 marks | 1MCQ
Routh-Hurwitz determines:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q7

Q49. Network, Signals & Systems | 2 marks | 2MSQ
In the circuit below, the opamp is ideal. $R_1 = C$, $R_2 = R$, $R_3 = R$. If the circuit is to show sustained oscillations, the respect...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q50

Q50. Network, Signals & Systems | 2 marks | 2MCQ
Consider a part of an electrical network as shown below. Some node voltages, and the current flowing through the 3 resis...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q19

Q51. Network, Signals & Systems | 2 marks | 2MCQ
For the two port network shown below, the value of the Y_{21} parameter (in Siemens) is _____. $I_1 I_2 1 + 2 + 4 2 - V_2$
 $V_1 + \dots$

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q60

Q52. Network, Signals & Systems | 2 marks | 2MCQ
A 10-bit analog-to-digital converter (ADC) has a sampling frequency of 1 MHz and a full scale voltage of 3.3 V. For an
I...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q44

Q53. Network, Signals & Systems | 2 marks | 2MCQ
In the circuit below, the opamp is ideal. $R_1 _ C R R + R C$ If the circuit is to show sustained oscillations, the respect...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q50

Q54. Network, Signals & Systems | 2 marks | 2NAT
Let F_1, F_2 , and F_3 be functions of (x, y, z) . Suppose that for every given pair of points and in space, the line integr...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q54

Q55. Network, Signals & Systems | 2 marks | 2MCQ
A source transmits a symbol s , taken from $\{-4, 0, 4\}$ with equal probability, over an additive white Gaussian noise
chann...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q45

Q56. Network, Signals & Systems | 2 marks | 2MCQ
In the circuit below, the opamp is ideal. $R_1 _ C R R + R C$ If the circuit is to show sustained oscillations, the respect...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q50

Q57. Network, Signals & Systems | 2 marks | 2NAT
A 4-bit priority encoder has inputs 3, 2, 1, and 0 in descending order of priority. The two-bit output is generated as 0...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q40

Q58. Electromagnetics | 1 marks | 1MCQ
Waveguide supports:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q7

Q59. Network, Signals & Systems | 2 marks | 2NAT
Which of the following statements involving contour integrals (evaluated counter- clockwise) on the unit circle in the
C...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q52

Q60. Network, Signals & Systems | 2 marks | 2MCQ
Consider a continuous-time finite-energy signal $x(t)$ whose Fourier transform vanishes outside the frequency interval $[-f_c, f_c]$, ...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q38

Q61. Network, Signals & Systems | 2 marks | 2MCQ
A uniform plane wave with electric field $E(x) = A e^{j(2\pi x - \omega t)}$ V/m is travelling in the air (relative permittivity, $\epsilon_r = 4$), ...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q39

Q62. Network, Signals & Systems | 2 marks | 2NAT
A digital communication system transmits through a noiseless bandlimited channel $[-B, B]$. The received signal $z(t)$ at the output is ...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 1 Q14

Q63. Network, Signals & Systems | 2 marks | 2MCQ
Consider a system where $x_1(t)$, $x_2(t)$, and $x_3(t)$ are three internal state signals and $x(t)$ is the input signal. The differential equation governing the system is ...

Answer: See explanation

Source: Adapted from GATE 2025 EC Session 1 Q53

Q64. Network, Signals & Systems | 1 marks | 1MCQ
Two identical sheets A and B, of dimensions 24 cm $\times$ 16 cm, can be folded into half using two distinct operations, FO1 and FO2, ...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 7 Q10

Q65. Network, Signals & Systems | 1 marks | 1MCQ
Four identical cylindrical chalk-sticks, each of radius = 0.5 cm and length = 10 cm, are bound tightly together using a string, ...

Answer: See explanation

Source: Adapted from GATE 2024 EC Session 7 Q7